

ELEAH BURMAN

New York, NY | (516) 965-3140 | EleahJBurman@gmail.com | GitHub | LinkedIn | Portfolio

Software engineer pursuing an M.S. in Computer Science (AI/ML) at Northeastern University, with production experience automating data pipelines in Python and Java and building full-stack applications with generative AI (Anthropic Claude API), React, and RESTful APIs. Comfortable across the stack from database design to UI, with hands-on exposure to AWS, automated testing, and Agile collaboration.

EXPERIENCE

Software Engineer, Tokio Marine HCC

June 2023 – Present

New York, NY

- Automate structured data pipelines using Python, Java, Camelot (PDF extraction), and Pandas, streamlining risk-management workflows for the insurance team.
- Build Java and Pandas automation that extracts and formats Excel tables for client-facing email reports, ensuring accurate, timely delivery of insurance data.
- Diagnose and resolve Python edge cases through root-cause analysis and long-term fixes, testing and maintaining production code in collaboration with the Risk Manager.

FEATURED PROJECTS

ClinicalNote AI

GitHub | Demo June 2026

Full-stack app (Python, TypeScript, FastAPI, Anthropic Claude API, React, Tailwind) using prompt engineering to convert unstructured clinical notes into structured summaries — patient presentation, mood assessment, diagnosis considerations, treatment and risk plans.

DisputeIQ

GitHub | Demo June 2026

End-to-end pipeline (Python, FastAPI, Claude API, LightGBM, React) that pairs LLM-based structured extraction with a gradient-boosted classifier to predict dispute outcomes with confidence scoring from unstructured text or PDF evidence.

Resume Keyword Matcher

GitHub | Demo June 2025

Flask application using NLTK and spaCy to score resume-job compatibility, with intelligent keyword extraction and fuzzy-matching for optimization recommendations.

Additional Projects

Budget Buddy — Node.js/React/MongoDB app for creating multi-currency budgets and tracking expenses, secured with cookie- and refresh-token-based auth. (GitHub)

Drag Queen Collector — Python/Django/PostgreSQL app on AWS for curating drag performer outfits and performances. (GitHub | Demo)

Bite Buddy — React/Node/MongoDB group project that calls an external API to build customized recipe boards. (GitHub | Demo)

Caffeinate-Me — MEAN-stack app with Google OAuth for adding, reviewing, and discovering local cafes. (GitHub | Demo)

Cat Vs Cat Lady — HTML/CSS/Canvas/JavaScript tic-tac-toe game playable against another person or an AI opponent. (GitHub | Demo)

DNInfo App — Java app that extracts and manages IP information from an XML database or API, reading/writing records in XML, JSON, and CSV via a GUI. (GitHub)

Skillzhunter App — Java app with a GUI that pulls job listings from an API and lets users retrieve/set records across XML, JSON, and CSV formats. (GitHub)

HACKATHONS

Computer Vision Hackathon — "Broadest Impact" award

New York, NY

Implemented YOLO (Ultralytics) and integrated FiftyOne for computer-vision-based tree/car detection across NYC to support environmental sustainability analysis.

Girls in Tech NYC Hackathon — 2nd Place

New York, NY

Led front-end development (Python, Django) of EcoNeighbor, an app enabling communities to identify and act on local environmental issues via an interactive map.

SKILLS

Languages & Frameworks: Python, Java, JavaScript, TypeScript, React.js, Node.js, Express.js, Vue.js, Flask, FastAPI, Django, SQL

Databases & Cloud: MongoDB, PostgreSQL, AWS

Tools & Practices: Git/GitHub, REST APIs, automated testing, Agile collaboration, generative AI / prompt engineering (Anthropic Claude API), Pandas, scikit-learn, LightGBM, Camelot

EDUCATION

Northeastern University — M.S. Computer Science (AI/ML focus)

Expected June 2027

General Assembly — Software Engineering Immersive (450+ hrs)

June 2023

New York University — B.F.A.